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Batch:A1

Experiment No. 6 – Advanced Parameterized Pipeline with Jenkins

The screenshot shows the Jenkins 'Configure' page for a job named 'Advanced-Jenkins-Pipeline'. The page is titled 'Configure' and has a sidebar with options: General (selected), Triggers, Pipeline, and Advanced. The main content area shows the 'General' tab with the following settings:

- ☐ Preserve stashes from completed builds ?
- ☒ This project is parameterized ?
- Choice Parameter ?**
 - Name ? : DEPLOY_ENV
 - Choices ? : development, staging, production
 - Description ? : Select deployment environment
 - Plain text [Preview](#)
 - [+ Add Parameter](#)
- ☐ Throttle builds ?

At the bottom, there is a 'Triggers' section and two buttons: 'Save' and 'Apply'.

Jenkins Code :

```
pipeline {
  agent any

  parameters {
    choice(
      name: 'DEPLOY_ENV',
      choices: ['development', 'staging', 'production'],
      description: 'Select deployment environment'
    )
  }

  environment {
    APP_NAME = 'StudentApp'
    VERSION = '1.0'
  }

  stages {

    stage('Information') {
      steps {
        echo "Application: ${APP_NAME}"
        echo "Version: ${VERSION}"
        echo "Environment: ${params.DEPLOY_ENV}"
        echo "Build Number: ${BUILD_NUMBER}"
      }
    }

    stage('Build') {
      steps {
        echo 'Building application...'
      }
    }
  }
}
```

```
}
```

```
stage('Test') {  
  steps {  
    echo 'Running tests...'  
  }  
}
```

```
stage('Deploy') {  
  when {  
    expression {  
      params.DEPLOY_ENV == 'staging'  
    }  
  }  
  
  steps {  
    echo 'Deploying application to STAGING...'  
  }  
}
```

```
stage('Production Approval') {  
  when {  
    expression {  
      params.DEPLOY_ENV == 'production'  
    }  
  }  
  
  steps {  
    input message: 'Approve production deployment?',  
           ok: 'Deploy'  
  }  
}
```

```
stage('Production Deploy') {
```

```
when {  
  expression {  
    params.DEPLOY_ENV == 'production'  
  }  
}  
  
steps {  
  echo 'Deploying application to PRODUCTION...'  
}  
}  
  
post {  
  success {  
    echo 'Pipeline completed successfully.'  
  }  
  
  failure {  
    echo 'Pipeline failed. Check the console output.'  
  }  
  
  always {  
    echo 'Pipeline execution completed.'  
  }  
}  
}
```

Staging:

Jenkins

Advanced-Jenkins-Pipeline

Status

Changes

Build with Parameters

Delete Pipeline

Configure

Rename

Stages

Pipeline Syntax

Builds >

No builds

Pipeline Advanced-Jenkins-Pipeline

This build requires parameters:

DEPLOY_ENV

Select deployment environment

development

development

staging

production

#1 - Advanced-Jenkins-Pipeline

Ask Gemini

localhost:8080/job/Advanced-Jenkins-Pipeline/1/stages/

Jenkins

Advanced-Jenkins-Pipeline

#1

Stages

#1

Started by admin

Started 1 min 6 sec ago

Took 7.1 sec

Parameters

Rerun

Settings

Start

Information

Build

Test

Deploy

Production Approval

Production Deploy

Post Actions

End

Search

Information 0.27s

Build 84ms

Test 80ms

Deploy 0.19s

Production Approval 0.25s

Production Deploy 83ms

Post Actions 0.14s

Post Actions

0.14s

Started 1m ago

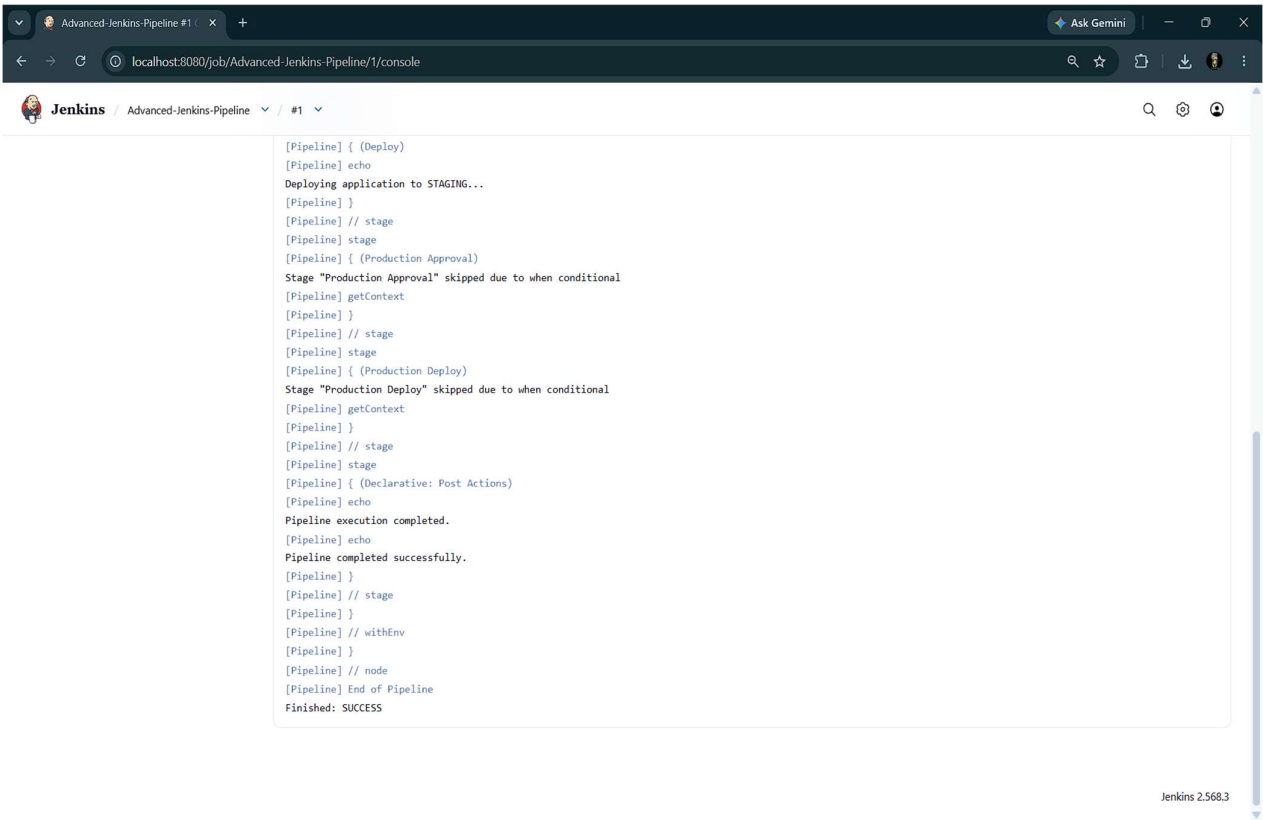
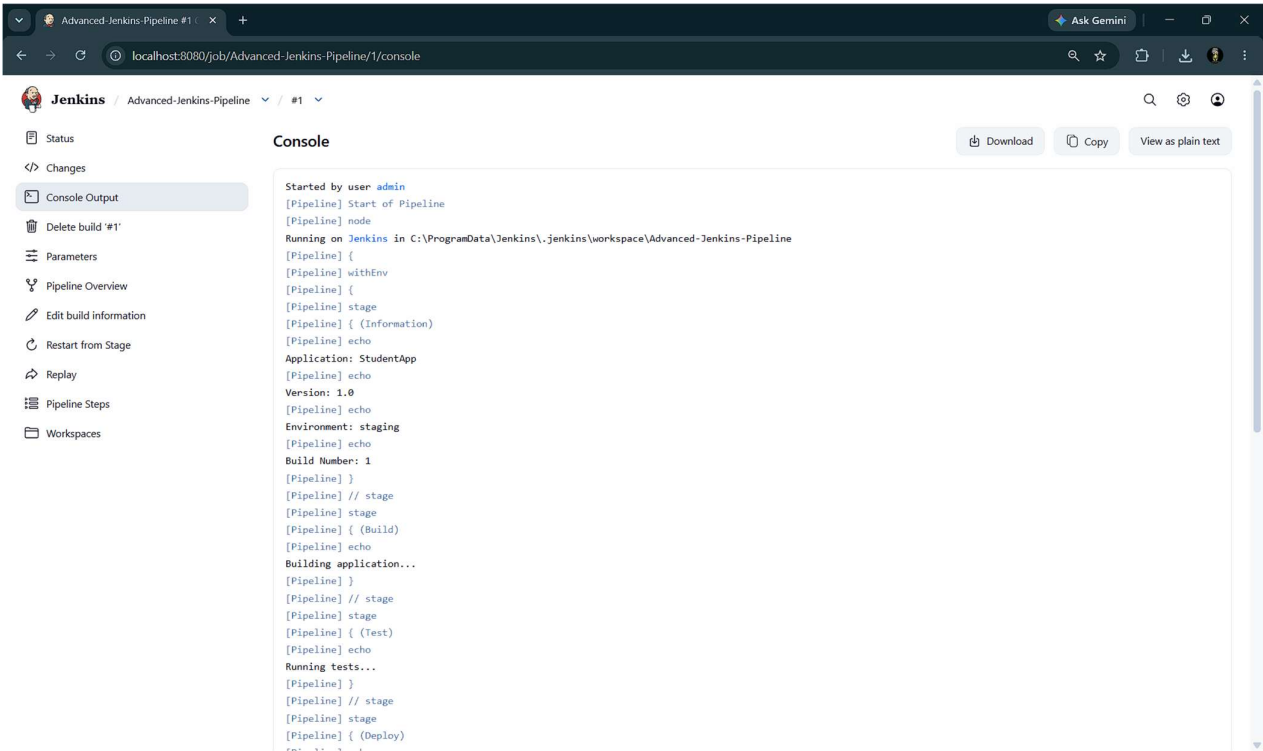
Jenkins

Pipeline execution completed. 52ms

Pipeline completed successfully. 35ms

Pipeline completed successfully.

Staging Console Output:



Productions:

#2 - Advanced-Jenkins-Pipeline

Ask Gemini

localhost:8080/job/Advanced-Jenkins-Pipeline/2/stages/

Jenkins / Advanced-Jenkins-Pipeline / #2 / Stages

< #2

Run

Settings

Started by admin

Started 33 sec ago

Build has been executing for 33 sec

Parameters

Start

Information

Build

Test

Deploy

Production Approval

Production Deploy

Post Actions

End

+

-

↺

Search

Post Actions

0.12s

Started <1s ago

Jenkins

...

Information 0.16s

Build 67ms

Test 68ms

Deploy 96ms

Production Approval 48s

Production Deploy 0.11s

Post Actions 0.12s

✓ Pipeline execution completed. 39ms

✓ Pipeline completed successfully. 30ms

0 Pipeline completed successfully.

Jenkins 2.568.3

Test Fail:

Advanced-Jenkins-Pipeline Co

Ask Gemini

localhost:8080/job/Advanced-Jenkins-Pipeline/configure

Jenkins / Advanced-Jenkins-Pipeline / Configure

Configure

General

Triggers

Pipeline

Advanced

Definition

Pipeline script

Script

29 steps {
30 echo 'Building application...'
31 }
32 }
33
34 stage('Test') {
35 steps {
36 echo 'Running tests...'
37 error 'Test failed!'
38 }
39 }
40
41 stage('Deploy') {
42 when {
43 expression {
44 }
45 }
46 }
47 }
48 }
49 }
50 }

Use Groovy Sandbox

Pipeline Syntax

Advanced

Advanced

Save

Apply

Test-Fail Console Output:

Advanced-Jenkins-Pipeline #3

Ask Gemini

localhost:8080/job/Advanced-Jenkins-Pipeline/3/console

Jenkins

Advanced-Jenkins-Pipeline

#3

Console Output

Delete build '#3'

Parameters

Pipeline Overview

Edit build information

Restart from Stage

Replay

Pipeline Steps

Workspaces

Previous Build

Started by user admin

[Pipeline] Start of Pipeline

[Pipeline] node

Running on Jenkins in C:\ProgramData\Jenkins\jenkins\workspace\Advanced-Jenkins-Pipeline

[Pipeline] {

[Pipeline] withEnv

[Pipeline] {

[Pipeline] stage

[Pipeline] { (Information)

[Pipeline] echo

Application: StudentApp

[Pipeline] echo

Version: 1.0

[Pipeline] echo

Environment: staging

[Pipeline] echo

Build Number: 3

[Pipeline] }

[Pipeline] // stage

[Pipeline] stage

[Pipeline] { (Build)

[Pipeline] echo

Building application...

[Pipeline] }

[Pipeline] // stage

[Pipeline] stage

[Pipeline] { (Test)

[Pipeline] echo

Running tests...

[Pipeline] error

[Pipeline] }

[Pipeline] // stage

[Pipeline] stage

[Pipeline] { (Deploy)

Stage "Deploy" skipped due to earlier failure(s)

[Pipeline] getContext

[Pipeline] }

Advanced-Jenkins-Pipeline #3

Ask Gemini

localhost:8080/job/Advanced-Jenkins-Pipeline/3/console

Jenkins

Advanced-Jenkins-Pipeline

#3

Stage "Deploy" skipped due to earlier failure(s)

[Pipeline] getContext

[Pipeline] }

[Pipeline] // stage

[Pipeline] stage

[Pipeline] { (Production Approval)

Stage "Production Approval" skipped due to earlier failure(s)

[Pipeline] getContext

[Pipeline] }

[Pipeline] // stage

[Pipeline] stage

[Pipeline] { (Production Deploy)

Stage "Production Deploy" skipped due to earlier failure(s)

[Pipeline] getContext

[Pipeline] }

[Pipeline] // stage

[Pipeline] stage

[Pipeline] { (Declarative: Post Actions)

[Pipeline] echo

Pipeline execution completed.

[Pipeline] echo

Pipeline failed. Check the console output.

[Pipeline] }

[Pipeline] // stage

[Pipeline] }

[Pipeline] // withEnv

[Pipeline] }

[Pipeline] // node

[Pipeline] End of Pipeline

ERROR: Test failed!

Finished: FAILURE

Jenkins 2.568.3

Build History:

Ask Gemini

localhost:8080/view/all/builds

Jenkins

All / Build History

+ New Item

Build History

Build Queue

No builds in the queue.

Build Executor Status

0/2

Build History of Jenkins

S	Build	Time Since	1	Status
	Advanced-Jenkins-Pipeline #3	3 min 28 sec		broken since this build
	Advanced-Jenkins-Pipeline #2	8 min 49 sec		stable
	Advanced-Jenkins-Pipeline #1	20 min		stable
	Assignment-CICD-Pipeline #1	1 hr 38 min		stable

Icons:

S M L

REST API

Jenkins 2.568.3

Update Information:

Advanced-Jenkins-Pipeline Con

Ask Gemini

localhost:8080/job/Advanced-Jenkins-Pipeline/configure

Jenkins

Advanced-Jenkins-Pipeline

Configure

Configure

General

Triggers

Pipeline

Advanced

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

```
18
19 stage('Information') {
20   steps {
21     echo "Application: ${APP_NAME}"
22     echo "Version: ${VERSION}"
23     echo "Environment: ${params.DEPLOY_ENV}"
24     echo "Build Number: ${BUILD_NUMBER}"
25     echo "Job Name: ${JOB_NAME}"
26     echo "Workspace: ${WORKSPACE}"
27     echo "Build URL: ${BUILD_URL}"
28   }
29 }
30
31 stage('Build') {
32   steps {
```

☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

Advanced

Advanced

Save

Apply

All - Jenkins

Ask Gemini

localhost:8080/view/all/builds

Jenkins

All

Build History

+ New Item

Build History

Build Queue

No builds in the queue.

Build Executor Status

0/2

Build History of Jenkins

S	Build	Time Since 1	Status
✓	Advanced-Jenkins-Pipeline #4	1 min 40 sec	back to normal
✗	Advanced-Jenkins-Pipeline #3	15 min	broken since this build
✓	Advanced-Jenkins-Pipeline #2	21 min	stable
✓	Advanced-Jenkins-Pipeline #1	32 min	stable
✓	Assignment-CICD-Pipeline #1	1 hr 51 min	stable

Icons: S M L

Conclusion:

The experiment successfully demonstrated the use of advanced Jenkins Pipeline features, including parameterized builds, environment variables, conditional stages, manual production approval, and post-build actions. The pipeline was executed for different deployment environments such as staging and production, and failure handling was also tested. Through this experiment, we gained a better understanding of Jenkins Pipeline automation, build status, console output, workspace, and build history. Overall, the experiment provided practical knowledge of using Jenkins for efficient and controlled CI/CD automation.